Assignment 2 – GitHub Collaboration & Automation (Report)

Field	Value
Group size	2
Chosen theme	Unit Converter
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Student ID	154097 / 146126
Course code & section	CSC3324 02
GitHub username	Syassine48
GitHub repository URL	https://github.com/Syassine48/unit--converter
GitHub Pages URL	https://syassine48.github.io/unit--converter/

1. Project & Role Distribution

The project is a small Python Unit Converter with three feature areas: length (`converter/length.py`), temperature (`converter/temperature.py`) and mass (`converter/mass.py`). I implemented the length and temperature features. Jihane did the mass feature.

2. Branches, Commits & Contributions

Each feature was built on its own branch with two commits: the code, then its unit tests.

Branch	Commits
`feature/length`	`c372e78` Add length conversion with unit table and validation; `751cbe5` Add unit tests for length conversion
`feature/temperature`	`435b4a9` Add temperature conversion with absolute zero validation; `51a03e4` Add unit tests for temperature conversion
`feature/mass`	`273b0f7` Add mass conversion with unit table and validation; `2e04584` Add unit tests for mass conversion
`docs/status-one`, `docs/status-two`	README status line commits used for the merge conflict (`fca78f0`, `66b9c8c`)

Branch	Commits
`ci/add-workflow`	GitHub Actions workflow and its fix
`docs/github-pages`, `docs/final-page-update`, `docs/add-report`	Project page, final page change and this report [confirm names]

3. Pull Requests & Code Review

Every change reached `main` through a Pull Request: #1 (`docs/status-one`), #2 (`docs/status-two`), the three feature PRs (#[n], #[n], #[n]), the CI workflow PR (#[n]), the Pages PRs (#[n], #[n]) and the report PR (#[n]).

4. Merge Conflict

The README contains the line `Project status: Development in progress.` We created two branches from the same version of `main`: `docs/status-one` changed it to "Project status: Core structure ready, features in review." and `docs/status-two` changed it to "Project status: Features under development, CI coming next." [confirm the exact wording you used]. We merged `docs/status-one` first through PR #1.

On `docs/status-two` we ran `git fetch origin` and `git merge origin/main`, which produced the conflict in `README.md`. Current (HEAD) was my `docs/status-two` version; Incoming (`origin/main`) was the version already merged from `docs/status-one`. We combined both into one line: "Project status: Core structure ready, features under development, CI coming next.", staged the file, committed (`66b9c8c` Resolve README status conflict by combining both versions), pushed, and merged PR #2.

5. Unit Tests

The suite has 9 pytest tests in `tests/test_length.py`, `tests/test_temperature.py` and `tests/test_mass.py`, all passing locally with `py -m pytest -v`.

Type	Examples
Normal behavior	2 km → 2000 m; 100 °C → 212 °F; 1 kg → 2.20462 lb
Edge / boundary	0 mi → 0 cm; 0 K → −273.15 °C (absolute zero); converting g to g returns the same value
Invalid input / error	unsupported unit (`parsec`) raises ValueError; −300 °C (below absolute zero) raises ValueError; negative mass raises ValueError

6. Continuous Integration

The workflow `.github/workflows/tests.yml` runs on every push and every Pull Request. It has one job, `test`, on an `ubuntu-latest` runner. Its steps are: check out the code, set up Python 3.12, install the dependencies with `pip install -r requirements.txt`, and run `pytest -v`.

The first run failed. The cause was a typo we introduced on purpose in the workflow: `pip install -r requirements.txtt` (an extra "t").

7. Ruleset & Protected main

The active ruleset `Protect main` targets `main` with an empty bypass list. It requires a Pull Request before merging, restricts branch deletion and blocks force pushes. After the workflow was running, we added the rule that the status check `test` must pass before merging.

8. Continuous Deployment

GitHub Pages is configured under Settings → Pages with the source "Deploy from a branch", branch `main`, folder `/docs`. Every merge to `main` that changes `docs/` triggers a "pages build and deployment" run, which updates the live site at https://syassine48.github.io/unit--converter/.

10. Fair Use of AI

we used Claude as an assistant.

11. Sources & Reflection

Sources: GitHub Docs: Actions, GitHub Docs: Pages, GitHub Docs: Rulesets, pytest documentation, and the course assignment handout.

Reflection: The hardest problem was that I first committed my length feature on `main` instead of on a feature branch. The ruleset blocked pushing it, so I moved the commits onto `feature/length` with a fast-forward merge and reset local `main` to `origin/main`. If a Pull Request, merge, test or workflow behaves unexpectedly, I would first check which branch I am on, then `git status` and `git log --oneline --graph --all`, and then the ruleset and the Actions log on GitHub.